
Chapter 4: Sequences and Series

A Level Pure Maths Notes · Rexponential

Why this chapter matters

A **sequence** is an ordered list of numbers (terms); a **series** is what you get when you add the terms of a sequence together. What's new in this chapter is the notation (particularly sigma notation), recurrence relations defined by a rule rather than a formula, and the **binomial expansion**, which needs real care, especially the negative/fractional-power version in section 4.6.

Throughout this chapter, keep two questions in mind for every problem: *is this arithmetic or geometric (or neither)?* and *am I being asked for a term, or a sum?* Mixing these up is the single most common source of lost marks.

4.1 Arithmetic sequences and series

Concept

An arithmetic sequence has a constant difference d between consecutive terms: each term is the last term plus d . This constant rate of change is exactly the discrete version of a straight line — if you plotted term number against term value, the points would lie on a line of gradient d . That's the conceptual link worth holding onto: arithmetic = linear growth.

Procedure

KEY FORMULAS

- n th term: $u_n = a + (n - 1)d$, where $a = u_1$ is the first term.
- Sum of the first n terms: $S_n = n/2 \cdot [2a + (n - 1)d] = n/2 \cdot (a + l)$, where l is the last term — the second form is just "average of first and last, times how many terms," which is quicker when you already know the last term.
- To find n when you're told a term's value, set u_n equal to that value and solve for n (it must come out as a positive integer).

Worked example

EXAMPLE 4.1

An arithmetic sequence has first term 5 and common difference 3. (a) Find the 20th term. (b) Find the sum of the first 20 terms. (c) Find the sum of all multiples of 7 between 100 and 300 (inclusive).

Step-by-step solution. (a) $u_{20} = a + 19d = 5 + 19(3) = 5 + 57 = \mathbf{62}$ (M1 for substituting into $u_n = a + (n-1)d$, A1 for 62).

(b) $S_{20} = 20/2 \cdot [2(5) + 19(3)] = 10 \cdot [10 + 57] = 10 \times 67 = \mathbf{670}$ (M1, A1).

(c) This is the type of question that trips people up because it's arithmetic *in disguise* — you first have to identify a , d , l , n yourself.

- The multiples of 7 form an arithmetic sequence with $d = 7$.
- First one ≥ 100 : $100 \div 7 = 14.28\dots$, so the first multiple is $15 \times 7 = \mathbf{105}$. (B1)
- Last one ≤ 300 : $300 \div 7 = 42.86\dots$, so the last multiple is $42 \times 7 = \mathbf{294}$. (B1)
- Number of terms: $294 = 105 + (n-1)(7) \Rightarrow 189 = 7(n-1) \Rightarrow n-1 = 27 \Rightarrow \mathbf{n = 28}$ (M1).
- $S_{28} = 28/2 \cdot (105 + 294) = 14 \times 399 = \mathbf{5586}$ (M1, A1).

EXAM TIP

Forgetting to round the endpoints *toward the inside* of the range (rounding $100 \div 7$ down instead of up, or $300 \div 7$ up instead of down) — always sanity-check that your first and last terms are actually multiples of d and actually lie inside the stated range.

4.2 Geometric sequences and series, sum to infinity

Concept

A geometric sequence has a constant *ratio* r between consecutive terms: each term is the last term times r . Where arithmetic is discrete linear growth, geometric is discrete **exponential** growth (or decay, if $|r| < 1$) — this is the link to compound interest, radioactive decay, and population models, and it's why geometric series matter so much in applications.

Procedure

KEY FORMULAS

- n th term: $u_n = ar^{n-1}$.
- Sum of the first n terms: $S_n = a(1 - r^n)/(1 - r)$ (equivalently $a(r^n - 1)/(r - 1)$ — use whichever avoids a negative divided by a negative).
- **Sum to infinity:** if $|r| < 1$, as $n \rightarrow \infty$ the term $r^n \rightarrow 0$, so the sum converges to $S_\infty = a/(1 - r)$. If $|r| \geq 1$ the series diverges and S_∞ does not exist — always state the condition $|r| < 1$ when you use this formula; it's an easy, cheap mark that's often missed.

Worked example

EXAMPLE 4.2

(a) A geometric sequence has first term 3 and common ratio 2. Find the 10th term and the sum of the first 10 terms. (b) A geometric sequence has first term 8 and common ratio $1/4$. Find the sum to infinity. (c) A geometric sequence has $u_1 = 6$ and $u_4 = 48$. Given the sequence has a positive common ratio, find r .

Step-by-step solution. (a) $u_{10} = 3 \times 2^9 = 3 \times 512 = \mathbf{1536}$. $S_{10} = 3(2^{10} - 1)/(2 - 1) = 3 \times 1023 = \mathbf{3069}$ (M1 for the formula, A1 each).

(b) Check $|r| = 1/4 < 1$, so the sum to infinity exists (B1 for stating this). $S_{\infty} = 8/(1 - 1/4) = 8/(3/4) = 8 \times 4/3 = \mathbf{32/3}$ (M1, A1).

(c) $u_4 = u_1 r^3$, so $48 = 6r^3 \implies r^3 = 8 \implies \mathbf{r = 2}$ (M1, A1 — remember $r^3 = 8$ only gives one real solution here since we're told r is positive; in general a cube root has one real solution anyway, unlike a square root, so no $\pm$ needed).

EXAM TIP

Using the sum-to-infinity formula without checking $|r| < 1$ — if a question gives $r = 3$, say, and asks for the "sum to infinity," the correct answer is that it *doesn't exist*, not a numerical value from blindly applying the formula.

4.3 Sigma notation

Concept

Sigma notation, Σ , is just compact shorthand for "add up a list of terms generated by a formula." $\sum_{r=1}^n f(r)$ means: substitute $r = 1, 2, 3, \dots, n$ into $f(r)$ in turn, and add up the results. It looks intimidating the first time you see it, but it's purely notation — it doesn't introduce any new mathematics, and every sigma sum in this course is either arithmetic or geometric once you expand it.

Procedure

- To **evaluate** $\sum_{r=1}^n f(r)$: work out the first few terms, identify whether they're arithmetic or geometric, find a , d (or r), and n , then apply the S_n formula from 4.1 or 4.2.
- To **write a sum in sigma notation**: spot the pattern in the terms (the common difference or ratio) and express the general term as a formula in r , then state the limits.
- Standard results worth knowing: $\sum_{r=1}^n 1 = n$, and more generally, splitting a sum like $\sum (ar + b)$ into $a\sum r + \sum b$ can simplify algebra-heavy sigma sums, though for this course direct evaluation via the AP/GP formulas is usually faster.

Worked example

EXAMPLE 4.3

(a) Evaluate $\sum_{r=1}^6 (3r - 2)$. (b) Write the sum $5 + 8 + 11 + \dots + 32$ using sigma notation, and evaluate it.

Step-by-step solution. (a) The terms are ($r=1$): 1, ($r=2$): 4, ($r=3$): 7, ... — an arithmetic sequence with $a = 1$, $d = 3$, $n = 6$ (M1 for identifying this). $S_6 = 6/2 \cdot [2(1) + 5(3)] = 3 \times 17 = \mathbf{51}$ (A1). (Check by direct addition: $1+4+7+10+13+16 = 51$ ✓.)

(b) The terms increase by 3 each time (arithmetic), so the general term is $u_r = 5 + (r-1)(3) = 3r + 2$ (M1). The last term $32 = 3r + 2 \implies r = 10$, so the sum is $\sum_{r=1}^{10} (3r + 2)$ (A1). Evaluating: $a = 5$, $d = 3$, $n = 10$, $S_{10} = 10/2 \cdot (5 + 32) = 5 \times 37 = \mathbf{185}$ (M1, A1).

EXAM TIP

Misreading the lower limit — sums written with $r = 1$ at the bottom of $\sum$ versus $r = 0$ at the bottom give different numbers of terms and a different first term; always write out the first one or two terms explicitly to check you've matched the formula to the notation correctly.

4.4 Recurrence relations

Concept

A recurrence relation defines each term using the *previous* term(s), e.g. $u_{n+1} = f(u_n)$, rather than giving a direct formula for u_n in terms of n . This is a different way of thinking about a sequence: you generate it step by step rather than reading off any term instantly. Recurrence relations can behave in three broad ways worth recognising by eye: **increasing** (terms keep growing), **decreasing** (terms keep falling), or **periodic** (the sequence repeats itself after a fixed number of steps, called the period).

Procedure

- Given u_1 and the rule $u_{n+1} = f(u_n)$, generate terms by direct substitution: $u_2 = f(u_1)$, $u_3 = f(u_2)$, and so on.
- To classify behaviour: compute several terms and look for a pattern. If $u_{n+1} > u_n$ throughout, the sequence is increasing (and similarly for decreasing). If the values start repeating exactly, find the smallest number of steps before repetition — that's the period.
- To **prove** increasing/decreasing algebraically (rather than just observing it), consider $u_{n+1} - u_n$ and show its sign is always the same.

Worked example

EXAMPLE 4.4

(a) A sequence is defined by $u_1 = 3$, $u_{n+1} = 2u_n - 5$. Find u_2 , u_3 , u_4 , u_5 and describe the sequence's behaviour. (b) A sequence is defined by $u_1 = 2$, $u_{n+1} = 1/(1 - u_n)$. Find u_2 , u_3 , u_4 and describe what you notice.

Step-by-step solution. (a) $u_2 = 2(3) - 5 = 1$. $u_3 = 2(1) - 5 = -3$. $u_4 = 2(-3) - 5 = -11$. $u_5 = 2(-11) - 5 = -27$ (M1 for correct method, A1 for all values: 1, -3, -11, -27). Each term is less than the one before, so the sequence is **decreasing** (in fact it decreases without bound) (A1).

(b) $u_2 = 1/(1-2) = -1$. $u_3 = 1/(1-(-1)) = 1/2$. $u_4 = 1/(1 - 1/2) = 2 = u_1$ (M1, A1 for values -1, 1/2, 2). Since $u_4 = u_1$, the sequence repeats from here on: 2, -1, 1/2, 2, -1, 1/2, ... — it is **periodic with period 3** (A1).

EXAM TIP

Assuming a sequence is periodic just because two terms happen to look similar, without checking that the *whole cycle* repeats — always confirm by generating at least one full extra period and checking every term matches.

4.5 Binomial expansion (positive integer powers)

Concept

The binomial expansion gives a formula for expanding $(a + b)^n$ without multiplying it out term by term — this is new territory, so go slowly here. The key conceptual point: the coefficients in the expansion are the entries of Pascal's triangle, and they count *how many ways* you can choose which terms contribute an a and which contribute a b when you multiply out $(a+b)(a+b)\cdots(a+b)$, n times. That combinatorial meaning is exactly what " n choose r ," written nC_r or $(n \ r)$, calculates.

Procedure

For a positive integer n :

$$(a + b)^n = \sum_{k=0}^n ({}^nC_k) a^{n-k} b^k$$

where $(n \text{ choose } k) = n! / [k!(n-k)!]$. In practice: write out the terms for $k = 0, 1, 2, \dots$ up to n , working out $(n \text{ choose } k)$, the power of a (which starts at n and decreases by 1 each term), and the power of b (which starts at 0 and increases by 1 each term) — the powers of a and b always add to n in every term, which is a good self-check.

For "find the first four terms" or "find the coefficient of x^3 " type questions, you only need to compute the relevant terms, not the whole expansion.

Worked example

EXAMPLE 4.5

Find the first four terms, in ascending powers of x , of the expansion of $(2 + 3x)^5$.

Step-by-step solution. Here $a = 2$, $b = 3x$, $n = 5$. Using $({}^5C_k)$ for $k = 0, 1, 2, 3$: 1, 5, 10, 10.

- $k=0$: $({}^5C_0) 2^5 (3x)^0 = 1 \times 32 \times 1 = \mathbf{32}$
- $k=1$: $({}^5C_1) 2^4 (3x)^1 = 5 \times 16 \times 3x = \mathbf{240x}$
- $k=2$: $({}^5C_2) 2^3 (3x)^2 = 10 \times 8 \times 9x^2 = \mathbf{720x^2}$
- $k=3$: $({}^5C_3) 2^2 (3x)^3 = 10 \times 4 \times 27x^3 = \mathbf{1080x^3}$

(M1 for using the binomial formula with correct a , b , n ; A1 for each of the four terms.)

So $(2 + 3x)^5 = 32 + 240x + 720x^2 + 1080x^3 + \dots$ (up to the x^5 term, which we weren't asked for).

EXAM TIP

Forgetting to raise a to a power in every term — students often correctly track the powers of b (since that's "the new thing," $3x$) but carelessly drop the a^{n-k} factor, or forget that a itself is being raised to a power (here 2^5 , 2^4 , etc.), not just 1.

4.6 Binomial expansion (negative and fractional powers)

Concept

When n is *not* a positive integer — negative, or a fraction — $(1 + x)^n$ can still be expanded as a series, but two things change fundamentally from 4.5: the expansion never terminates (it goes on forever, term after term), and it is only **valid** — meaning it actually converges to $(1+x)^n$ — for $|x| < 1$. Outside that range the series doesn't represent the function at all. This validity condition is not optional exam small print; examiners specifically test whether you state it.

Procedure

For any n (not just positive integers):

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \dots \quad (|x| < 1)$$

If you're expanding $(a + bx)^n$ rather than $(1 + x)^n$, first factor out a : $(a + bx)^n = a^n(1 + (b/a)x)^n$, expand the bracket using the formula above with x replaced by $(b/a)x$, then multiply back through by a^n . The validity condition becomes $|(b/a)x| < 1$, i.e. $|x| < |a/b|$.

Worked example

EXAMPLE 4.6

Expand $(1 - 2x)^{-3}$ in ascending powers of x , up to and including the x^3 term, stating the range of values of x for which the expansion is valid.

Step-by-step solution. Here the expansion is directly in the form $(1 + X)^n$ with $X = -2x$ and $n = -3$ (M1 for identifying n and X correctly — this is the step where sign errors creep in).

- Constant term: 1
- x^1 term: $n \cdot X = (-3)(-2x) = 6x$
- x^2 term: $[n(n-1)/2!] X^2 = [(-3)(-4)/2] (-2x)^2 = 6 \times 4x^2 = 24x^2$
- x^3 term: $[n(n-1)(n-2)/3!] X^3 = [(-3)(-4)(-5)/6] (-2x)^3 = (-10)(-8x^3) = 80x^3$

So $(1 - 2x)^{-3} = 1 + 6x + 24x^2 + 80x^3 + \dots$ (A1 for each of the three coefficients $6x$, $24x^2$ and $80x^3$ — the constant term is not separately credited)

Validity: we need $|X| < 1$, i.e. $|-2x| < 1$, i.e. $|x| < 1/2$, so the expansion is valid for $-1/2 < x < 1/2$ (M1, A1 — this mark is very commonly dropped even when the expansion itself is correct).

EXAM TIP

Losing track of signs when X itself is negative (here $X = -2x$): every odd-numbered term in the bracket $[n(n-1)(n-2)\dots]$ picks up its own sign from n being negative, *and* every odd power of X contributes another sign — work through each factor individually rather than trying to do it in your head, and don't forget the validity condition at the end.

Chapter summary

Key formulas

KEY FORMULAS

- **Arithmetic** — $u_n = a + (n-1)d$, $S_n = n/2[2a + (n-1)d] = n/2(a + l)$
- **Geometric** — $u_n = ar^{n-1}$, $S_n = a(1 - r^n)/(1 - r)$, $S_\infty = a/(1 - r)$ ($|r| < 1$)
- **Binomial (positive integer n)** — $(a + b)^n = \sum_{k=0}^n {}^nC_k a^{n-k} b^k$
- **Binomial (any n)** — $(1 + x)^n = 1 + nx + n(n-1)/2! x^2 + n(n-1)(n-2)/3! x^3 + \dots$ ($|x| < 1$)

Common exam mistakes across this chapter

- Confusing arithmetic (linear) with geometric (exponential) growth — always check whether consecutive terms have a constant *difference* or a constant *ratio* before choosing a formula.
- Applying S_∞ without checking $|r| < 1$.
- In sigma notation questions, misreading the limits and getting the wrong number of terms.
- In binomial expansions with n not a positive integer, forgetting the validity condition — this is a guaranteed mark on almost every exam question of this type, so make stating it a habit, not an afterthought.

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- Sign errors in the $(1+x)^n$ expansion when x itself carries a coefficient or a minus sign — substitute carefully and work one factor at a time.

You should now be able to

- Apply the arithmetic and geometric sequence/series formulas, including sum to infinity
- Evaluate a sigma-notation sum and write a given sum in sigma notation
- Generate terms from a recurrence relation and classify its behaviour (increasing, decreasing, periodic)
- Expand $(a+b)^n$ for positive integer n , and $(1+x)^n$ for negative or fractional n , stating the validity condition